

Mozhgan Nasr Azadani

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 [Mozhgan Nasr](#) |  [Personal Website](#) |  [Google Scholar](#)

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RESEARCH INTEREST

The goal of my research is to develop the algorithmic and representational foundations of multimodal AI systems that integrate vision, language, and action to enable grounded reasoning, predictive world modeling, and reliable decision-making in complex real-world environments.

EDUCATION

- **Ph.D. in Computer Science** Jan 2019 - Dec 2023
University of Ottawa Ottawa, Canada
 - Advisor: Prof. Azzedine Boukerche
 - Thesis title: "[Driving Behavior Analysis and Prediction for Safe Autonomous Vehicles](#)"
- **M.S. in Software Engineering** Sep 2015 - Jan 2018
Isfahan University of Technology Isfahan, Iran
 - Advisor: Prof. Nasser Ghadiri
 - Thesis title: "Graph-based biomedical text summarization"
- **B.S. in Computer Science** Sep 2009 - Jan 2013
University of Isfahan Isfahan, Iran

RESEARCH EXPERIENCE

- **Stanford University** Feb 2026 - Present
Visiting Postdoctoral Scholar Stanford, USA
 - Advisor: Prof. Marco Pavone at [ASL Lab](#).
 - My research focuses on grounded multimodal large language models and hallucination mitigation to improve the reliability and interpretability of multimodal reasoning. Duties also include mentoring students.
- **University of Waterloo** May 2024 - Present
Postdoctoral Fellow Waterloo, Canada
 - Advisor: Prof. Krzysztof Czarnecki at [WISE Lab](#).
 - My research focuses on multimodal large language models for autonomous driving. Duties also include mentoring PhD students, delivering guest lectures, grant proposal writing, and sub-reviewing grant proposals and academic papers.
- **University of Ottawa** Jan 2019 - May 2024
Graduate Research Assistant Ottawa, Canada
 - Advisor: Prof. Azzedine Boukerche at PARADISE Lab.
 - Devised a transformer-based algorithm for multimodal motion forecasting under uncertainty.
 - Designed a two-step multimodal goal and trajectory forecasting framework, validated on the Argoverse 1.
 - Created a context-aware, multi-agent motion forecasting model for AV using inverse reinforcement learning (IRL).
 - Created interaction-aware motion forecasting models based on Spatio-Temporal Attention Graph Networks.
 - Designed a multimodal motion planning and trajectory prediction system using Graph Neural Networks.
 - Developed a temporal neural network framework for driver identification through learned driver embeddings.
 - Built a Siamese neural network framework for driver verification and impostor detection.
- **University of Isfahan** June 2017 - Sep 2017
Summer Research Intern Isfahan, Iran
 - Collaborated with a multidisciplinary research team to develop employee health management software and assist in defining the project's research agenda.
 - Designed and implemented a machine learning framework for summarizing Electronic Medical Records (EMRs) using UMLS and the FP-Growth algorithm.
- **Isfahan University of Technology** Sep. 2015 - Jan. 2018
Graduate Research Assistant Isfahan, Iran
 - Designed a graph-based framework for summarizing Electronic Medical Records, achieving improved performance on the ROUGE evaluation metric.

HONORS AND AWARDS

- **Globalink Research Award** Feb. 2026 - Aug. 2026
Mitacs
 - Research award supporting the international research collaboration.
- **Postdoctoral Fellowship** 2024 - 2026
Natural Sciences and Engineering Research Council of Canada (NSERC)
 - Prestigious fellowship supporting outstanding postdoctoral researchers.
- **Merit Scholarship** 2021 – 2023
University of Ottawa
 - Awarded in recognition of outstanding academic achievement during doctoral studies.
- **International Doctoral Scholarship** 2019 – 2023
University of Ottawa
 - Competitive scholarship awarded to outstanding international Ph.D. students.
- **Excellence in Graduate Studies** 2018
Isfahan University of Technology
 - Awarded in recognition of exceptional graduate academic performance.

TEACHING

- **Autonomous Vehicles** Winter 2025
Guest Lecturer, University of Waterloo
 - Delivered a guest lecture titled *MLLMs: From Foundations to Driving with Natural Language*. 
- **Principles of Intelligent Transportation Systems (CSI 5107)** Winter 2025
Guest Lecturer, University of Ottawa
 - Delivered a guest lecture titled *Driving Behavior Analysis Guidelines for Intelligent Transportation Systems*.
- **Privacy Enhancing Technologies (CSI 5139)** Winter 2023
Teaching Assistant, University of Ottawa
- **Introduction to Formal Languages (CSI 3104) (×3)** Spring 2020 - Winter 2023
Teaching Assistant, University of Ottawa
- **Cryptography (CSI 4108 / SEC 5101) (×4)** Fall 2019 - Fall 2022
Teaching Assistant, University of Ottawa
- **Natural Language Processing (CSI 5386)** Winter 2022
Teaching Assistant, University of Ottawa
- **Data Mining and Concept Learning (CSI 5387)** Winter 2020
Teaching Assistant, University of Ottawa
- **Advanced Database Systems** Winter 2016
Teaching Assistant, Isfahan University of Technology
- **Database Systems II** Fall 2016
Teaching Assistant, Isfahan University of Technology

RESEARCH MENTORSHIP

Provided research/project mentorship to master's and Ph.D. students in multimodal foundation models, machine learning, computer vision, and robotics.

- 1 × M.Sc., 2 × Ph.D., Stanford University Feb. 2026 - Present
- 5 × M.Sc., 2 × Ph.D., University of Waterloo Dec. 2024 - Present
- 3 × M.Sc., University of Ottawa 2021 - 2024

PUBLICATIONS

For a complete and the most up-to-date publication record, please visit my [Google Scholar](#).

Preprints/Under reviews

- [P1] **M. Nasr Azadani**, Yimu Wang, Yongpeng Zhu, Lihong Chen, Milan Ganai, Sean Sedwards, Marco Pavone, Krzysztof Czarnecki. [VistaQA: Benchmarking Joint Visual Question Answering and Pixel-Level Evidence](#). *Under review*, 2026.
- [P2] Jadelynn Dao*, Milan Ganai*, Yasmina Abukhadra, Ajay Sridhar, **M. Nasr Azadani**, Katie Luo, Clark Barrett, Jiajun Wu, Chelsea Finn, Marco Pavone. [DIRECT: When and Where Should You Allocate Test-Time Compute in Embodied Planners?](#). *Under review*, 2026.

- [P3] Yimu Wang, Yee Man Choi, Barry Zhang, **M. Nasr Azadani**, Sean Sedwards, Krzysztof Czarnecki. [Where Does the Answer Come From? Benchmarking View-Level Visual Evidence Identification in Multi-View MLLMs for Autonomous Driving](#). *Under review*, 2026.

Journal Articles

- [J1] **M. Nasr Azadani**, J. Riddell, S. Sedwards, and K. Czarnecki. [Rethinking the Mixture of Vision Encoders Paradigm for Enhanced Visual Understanding in Multimodal LLMs](#). *Transactions on Machine Learning Research (TMLR) (J2C certification)*, 2026.
- [J2] **M. Nasr Azadani**, and A. Boukerche. [Hierarchical Transformers for Motion Forecasting Based on Inverse Reinforcement Learning](#). *IEEE Transactions on Vehicular Technology (T-VT)*, 2025.
- [J3] **M. Nasr Azadani**, and A. Boukerche. [CAPHA: A Novel Context-Aware Behavior Prediction System of Heterogeneous Agents for Autonomous Vehicles](#). *IEEE Transactions on Vehicular Technology (T-VT)*, 2023.
- [J4] **M. Nasr Azadani**, and A. Boukerche. [STAG: A Novel Interaction-Aware Path Prediction Method Based on Spatio-Temporal Attention Graphs for Connected Automated Vehicles](#). *Ad Hoc Networks*, 2023.
- [J5] **M. Nasr Azadani**, and A. Boukerche. [A Novel Multimodal Vehicle Path Prediction Method Based on Temporal Convolutional Networks](#). *IEEE Transactions on Intelligent Transportation Systems (T-ITS)*, 2022.
- [J6] **M. Nasr Azadani**, and A. Boukerche. [Siamese Temporal Convolutional Networks for Driver Identification Using Driver Steering Behavior Analysis](#). *IEEE Transactions on Intelligent Transportation Systems (T-ITS)*, 2022.
- [J7] **M. Nasr Azadani**, and A. Boukerche. [DriverRep: Driver Identification Through Driving Behavior Embeddings](#). *Journal of Parallel and Distributed Computing*, 2022.
- [J8] **M. Nasr Azadani**, and A. Boukerche. [Driving Behavior Analysis Guidelines for Intelligent Transportation Systems](#). *IEEE Transactions on Intelligent Transportation Systems (T-ITS)*, 2021.
- [J9] **M. Nasr Azadani**, N. Ghadiri, and E. Davoodijam. [Graph-Based Biomedical Text Summarization: An Itemset Mining and Sentence Clustering Approach](#). *Journal of Biomedical Informatics*, 2018.

Conference Proceedings

- [C1] Y. Wang, **M. Nasr Azadani**, S. Sedwards, and K. Czarnecki. [HAWAII: Hierarchical Visual Knowledge Transfer for Efficient Vision-Language Models](#). *NeurIPS*, 2025.
- [C2] Y. Wang*, **M. Nasr Azadani***, S. Sedwards, and K. Czarnecki. [LEO-MINI: An Efficient Multimodal Large Language Model Using Conditional Token Reduction and Mixture of Multi-Modal Experts](#). *EMNLP*, 2025.
- [C3] **M. Nasr Azadani**, and A. Boukerche. [A Novel Transformer-Based Model for Motion Forecasting in Connected Automated Vehicles](#). *IEEE International Conference on Communications (ICC)*, 2024.
- [C4] **M. Nasr Azadani**, and A. Boukerche. [GMP: Goal-Based Multimodal Motion Prediction for Automated Vehicles](#). *IEEE Global Communications Conference (GLOBECOM)*, 2023.
- [C5] **M. Nasr Azadani**, and A. Boukerche. [A Context-Aware Path Forecasting Method for Connected Autonomous Vehicles](#). *IEEE International Conference on Communications (ICC)*, 2023.
- [C6] **M. Nasr Azadani**, and A. Boukerche. [A Social-Aware Vehicle Path Forecasting Method Using Graph Neural Networks](#). *ACM Symposium on Design and Analysis of Intelligent Vehicular Networks and Applications*, 2023.
- [C7] **M. Nasr Azadani**, and A. Boukerche. [A Novel Multimodal Behavior Prediction Method for Automated Vehicles](#). *ACM Symposium on Design and Analysis of Intelligent Vehicular Networks and Applications*, 2023.
- [C8] **M. Nasr Azadani**, and A. Boukerche. [An Interaction-Aware Vehicle Behavior Prediction for Connected Automated Vehicles](#). *IEEE International Conference on Communications (ICC)*, 2022.
- [C9] **M. Nasr Azadani**, and A. Boukerche. [An Internet-of-Vehicles Powered Defensive Driving Warning Approach for Traffic Safety](#). *IEEE Global Communications Conference (GLOBECOM)*, 2021.
- [C10] **M. Nasr Azadani**, and A. Boukerche. [Toward Driver Intention Prediction for Intelligent Vehicles: A Deep Learning Approach](#). *IEEE Conference on Local Computer Networks (LCN)*, 2021.
- [C11] **M. Nasr Azadani**, and A. Boukerche. [Driver Identification Using Vehicular Sensing Data: A Deep Learning Approach](#). *IEEE Wireless Communications and Networking Conference (WCNC)*, 2021.
- [C12] **M. Nasr Azadani**, and A. Boukerche. [Performance Evaluation of Driving Behavior Identification Models Through CAN-BUS Data](#). *IEEE Wireless Communications and Networking Conference (WCNC)*, 2020.
- [C13] **M. Nasr Azadani**, and N. Ghadiri. [Evaluating Different Similarity Measures for Automatic Biomedical Text Summarization](#). *Intelligent Systems Design and Applications*, 2017.
- [C14] E. Maleki*, **M. Nasr Azadani***, and N. Ghadiri. [Performance Evaluation of SpatialHadoop for Big Web Mapping Data](#). *International Conference on Web Research*, 2016.

Workshop Papers

[W1] M. Nasr Azadani, and A. Boukerche. [Convolutional and Recurrent Neural Networks for Driver Identification: An Empirical Study](#). *IEEE NOMS Workshop on Intelligent Transportation and Autonomous Vehicle Technologies*, 2022.

INVITED TALKS

- | | |
|---|-----------------|
| • Mixture of Vision Encoders for MLLMs
<i>Stanford University, CA, USA</i> | <i>Mar 2026</i> |
| • Toward Efficient and Reliable VLMs for Real-World Autonomous Systems
<i>Workshop on Embodied and Safe-Assured Robotic Systems, NeurIPS 2025</i>  | <i>Nov 2025</i> |
| • Driving Behavior Analysis Guidelines for Intelligent Transportation Systems
<i>Principle of Intelligent Transportation Systems, University of Ottawa, ON, Canada</i> | <i>Feb 2023</i> |
| • Graph-based Biomedical Text Summarization
<i>Isfahan University of Technology, Isfahan, Iran</i> | <i>Nov 2017</i> |

WORKSHOP ORGANIZATION

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|---|-----------------|
| • 2nd workshop on Grounded and Faithful Vision-Language Models for Real-World Deployment
<i>NeurIPS 2026 VLM4RWD (lead organizer)</i> <ul style="list-style-type: none">◦ Focused on grounded understanding and faithful reasoning of VLMs/VLAs | <i>Dec 2026</i> |
| • 1st workshop on Grounding Language Models: Learning Faithfully and Efficiently
<i>EMNLP 2026 GroundLM</i> <ul style="list-style-type: none">◦ Focused on grounding, faithfulness, and efficiency in large language models | <i>Oct 2026</i> |
| • 1st workshop on how to Build Effective World Models for Embodied AI
<i>ECCV 2026 WMEAI</i> <ul style="list-style-type: none">◦ Focused on world models integration into embodied AI systems | <i>Sep 2026</i> |
| • 1st workshop on Vision Language Models: Challenges of Real World Deployment
<i>NeurIPS 2025 VLM4RWD (lead organizer)</i> <ul style="list-style-type: none">◦ Focused on efficiency, robustness, and reliability of vision language models | <i>Nov 2025</i> |

ACADEMIC SERVICE

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| • Area Chair
<i>NeurIPS 2026</i> | <i>2026</i> |
| • Journal Reviewer
<i>I am an active reviewer for the following venues:</i> <ul style="list-style-type: none">◦ Transactions on Machine Learning Research (TMLR)◦ IEEE Transactions on Vehicular Technology (T-VT)◦ IEEE Transactions on Intelligent Vehicles (T-IV)◦ IEEE Transactions on Intelligent Transportation Systems (T-ITS)◦ Nature Communications◦ IEEE Transactions on Multimedia◦ Pattern Recognition◦ IEEE Access◦ IEEE Transactions on Mobile Computing◦ IEEE Transactions on Consumer Electronics◦ Applied Artificial Intelligence◦ The Open Transportation Journal | <i>2022 - present</i> |
| • Conference Reviewer
<i>I have served as a reviewer for the following venues:</i> <ul style="list-style-type: none">◦ IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)◦ IEEE/CVF International Conference on Computer Vision (ICCV)◦ Annual Meeting of the Association for Computational Linguistics (ACL)◦ Conference on Neural Information Processing Systems (NeurIPS)◦ IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)◦ IEEE International Conference on Communications (ICC)◦ IEEE Wireless Communications and Networking Conference (WCNC)◦ IEEE Global Communications Conference (GLOBECOM)◦ IEEE Intelligent Vehicles Symposium (IV)◦ IEEE International Conference on Distributed Smart Cameras (DSC) | <i>2023 - present</i> |

OTHER ACTIVITIES

- **IEEE Women in Engineering (WIE) Member**

2021 – Present

IEEE

- Participated in professional development, networking, and leadership events, including the IEEE WIE Leaders Forum organized by IEEE Region 7 and the Ottawa Section WIE.

- **Iranian Student Association of University of Ottawa**

2020 – 2022

Volunteer

- Organized and delivered webinars and orientation sessions for incoming students.